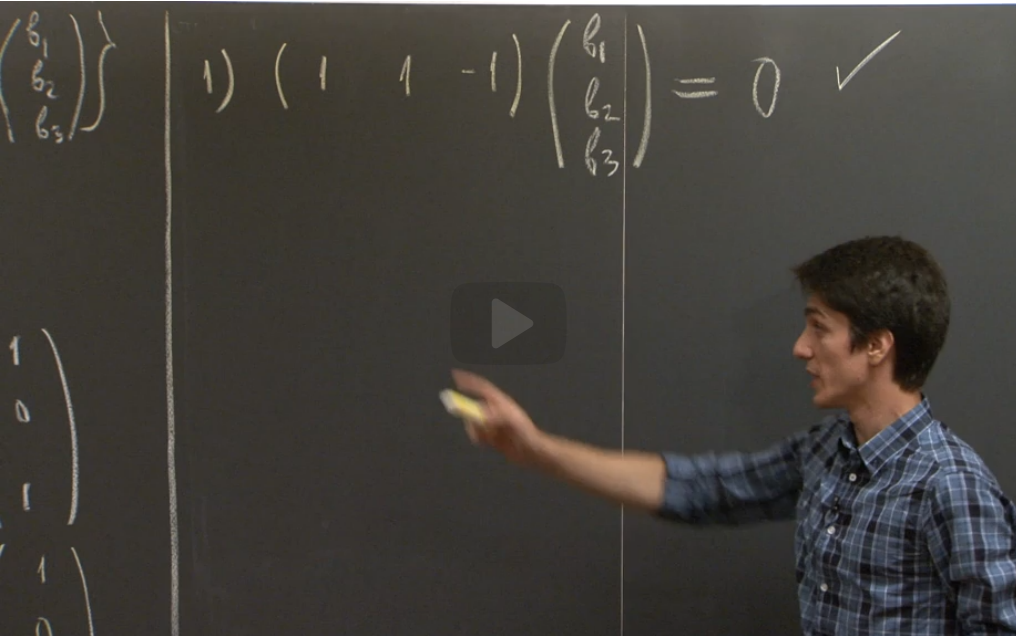


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Upgrade

Lecture due Jan 24, 2024 20:30 IST

Worked examples



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this as a matrix equation.
So in particular, b_1 plus b_2 minus b_3 equals 0.
We can just write it as the matrix 1, 1, negative 1, times b_1 , b_2 , b_3 equal to 0.
So therefore, b_1 , b_2 , and b_3 precisely describe the null space of 1, 1, negative 1.
And thus, what we are given in the first question
is a subspace.
What about the second one?
Again, we are given a relation between b_1 , b_2 , b_3 , but it's not linear.
 b_3 is the product of b_1 and b_2 .
So your guts tell you that this shouldn't be a vector space, but we need to prove why

Identify the subspaces

1/1 point (graded)
Which of the following subsets of $\mathbb{R}^2$ are subspaces? Check all that apply.

- ☐ The set of all vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ satisfying $x^2 + y^2 = 1$.
- ☐ The set of all vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ satisfying $xy = 0$.
- ☒ The set of all vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ satisfying $2x + 3y = 0$.
- ☐ None of these.



Solution:

Only the set of vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ satisfying $2x + 3y = 0$ is a subspace.

Algebraic explanation: Let S be the set. For S to be a vector space, it must satisfy all three conditions in the definition.

- The set of all vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ satisfying $x^2 + y^2 = 1$ doesn't even satisfy the first condition, because the zero vector $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is not in S .
- The set of all vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ satisfying $xy = 0$ satisfies the first condition: the zero vector is in S . It satisfies the second condition too: If $\begin{pmatrix} x \\ y \end{pmatrix}$ is one vector in S (so $xy = 0$) and c is any scalar, then the vector

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$c \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} cx \\ cy \end{pmatrix}$ satisfies

$$(cx)(cy) = c^2xy = c^2(0) = 0,$$

so $c \begin{pmatrix} x \\ y \end{pmatrix}$ is in S .

However, it does not satisfy the third condition for **some** pairs of vectors in S : for example, $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$ are in S , but their sum $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ is not in S .

- The set of all vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ satisfying $2x + 3y = 0$ is a vector space, as we will now check. First, the zero vector is in S . Second, if $\begin{pmatrix} x \\ y \end{pmatrix}$ is any element of S (so $2x + 3y = 0$) and c is any scalar, then multiplying the equation by c gives

$$2(cx) + 3(cy) = 0,$$

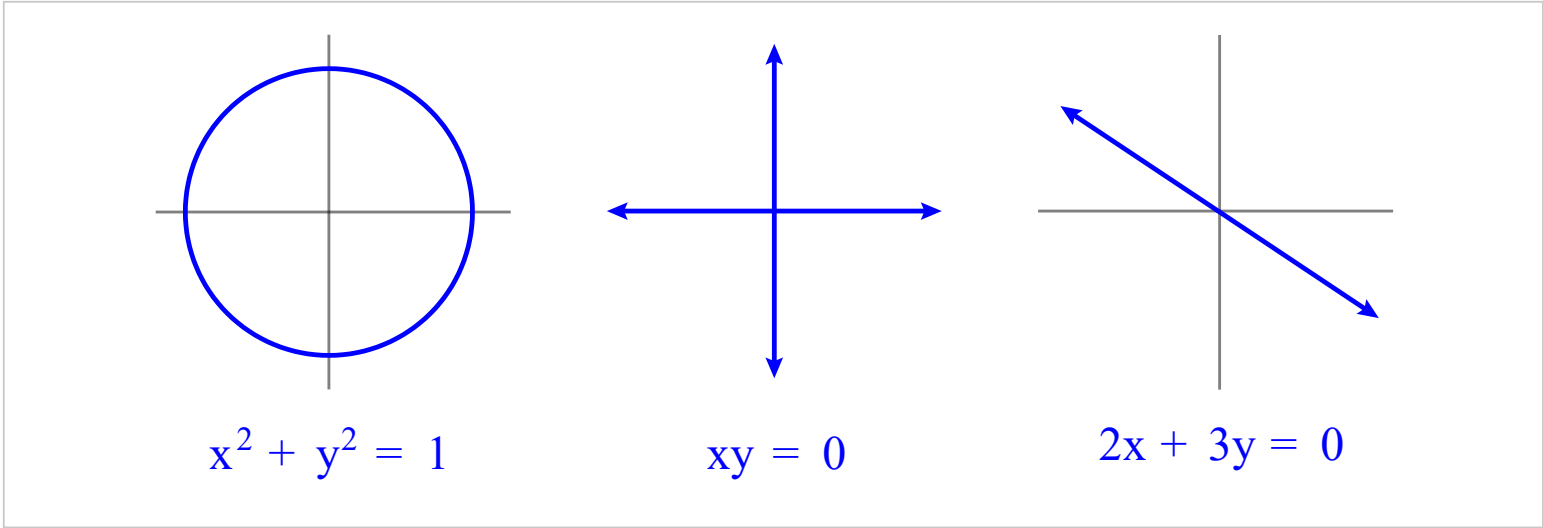
which shows that the vector $c \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} cx \\ cy \end{pmatrix}$ is in S . Third, if $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ and $\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ are in S (so $2x_1 + 3y_1 = 0$ and $2x_2 + 3y_2 = 0$), then adding the equations shows that

$$2(x_1 + x_2) + 3(y_1 + y_2) = 0,$$

which says that the vector

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}$$

is in S . Thus S is a vector space.



Alternate solution: We could have also solved this example using the fact that the only subspaces of $\mathbb{R}^2$ are zero, lines through the origin, and all of $\mathbb{R}^2$.

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Identify the subspaces II

1/1 point (graded)
Which of the following are subspaces of $\mathbb{R}^2$? Check all that apply.

☒ $y = x$

☒ $\{0\}$

☐ $y = 2x + 3$

☐ $y = 2$

☒ $x = 0$

☐ $y = 1/x$

☒ $y = -4x$

☐ $y = -x^2 + 1$

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- the point $\mathbf{0}$;
- any line passing through the origin.
- the entire plane $\mathbb{R}^2$.

The curves $y = 1/x$, $y = -x^2 + 1$, and $y = x^2$ are not subspaces of $\mathbb{R}^2$: both $y = 1/x$ and $y = -x^2 + 1$ fail to contain the origin $\mathbf{0}$, and $y = x^2$ fails to be closed under addition and scalar multiplication.

The lines $y = 2$ and $y = 2x + 3$ do not contain the zero vector, so they cannot be subspaces of $\mathbb{R}^2$.

The set $\{\mathbf{0}\}$ is a subspace of $\mathbb{R}^2$ by itself.

The lines $y = x$, $x = 0$ and $y = -4x$ are all subspaces because they all pass through the origin, and are closed under addition and scalar multiplication:

- The line $y = x$ is the set of vectors $\left\{\begin{pmatrix} a \\ a \end{pmatrix} \text{ where } a \text{ is real}\right\}$. Note that $\mathbf{0}$ is in this set, $c \begin{pmatrix} a \\ a \end{pmatrix} = \begin{pmatrix} ca \\ ca \end{pmatrix}$ remains in this set, and $\begin{pmatrix} a \\ a \end{pmatrix} + \begin{pmatrix} b \\ b \end{pmatrix} = \begin{pmatrix} a+b \\ a+b \end{pmatrix}$ also remains in the set.
- The line $x = 0$ is the set of vectors $\left\{\begin{pmatrix} 0 \\ a \end{pmatrix} \text{ where } a \text{ is real}\right\}$. Note that $\mathbf{0}$ is in this set, $c \begin{pmatrix} 0 \\ a \end{pmatrix} = \begin{pmatrix} 0 \\ ca \end{pmatrix}$ remains in this set, and $\begin{pmatrix} 0 \\ a \end{pmatrix} + \begin{pmatrix} 0 \\ b \end{pmatrix} = \begin{pmatrix} 0 \\ a+b \end{pmatrix}$ also remains in the set.
- The line $y = -4x$ is the set of vectors $\left\{\begin{pmatrix} a \\ -4a \end{pmatrix} \text{ where } a \text{ is real}\right\}$. Note that $\mathbf{0}$ is in this set, $c \begin{pmatrix} a \\ -4a \end{pmatrix} = \begin{pmatrix} ca \\ -4ca \end{pmatrix}$ remains in this set, and $\begin{pmatrix} a \\ -4a \end{pmatrix} + \begin{pmatrix} b \\ -4b \end{pmatrix} = \begin{pmatrix} a+b \\ -4(a+b) \end{pmatrix}$ also remains in the set.

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9. More practice with subspaces using nullspace and span

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